

# Math 115A — Discussion 01

Worksheet: Fields

Thursday, September 24, 2026

**Instructions.** Work in pairs. The field axioms (F1)–(F10) from the notes are your only tools in Part A — do not use any “obvious” algebraic fact without deriving it from those axioms first. Write complete sentences; every step should cite the axiom or earlier result that justifies it.

## Part A Consequences of the Axioms

Throughout,  $\mathbb{F}$  is a field with additive identity 0 and multiplicative identity 1. Recall from the notes:  $-a$  denotes the unique element with  $a + (-a) = 0$  (the *opposite* of  $a$ ; see Remark 1.1.3), and  $a^{-1}$  denotes the unique element with  $a \cdot a^{-1} = 1$  for  $a \neq 0$  (Exercise 1.1.5).

**Problem 1.** Let  $a \in \mathbb{F}$ . Prove that  $a \cdot 0 = 0$ .

**Problem 2.** Let  $a \in \mathbb{F}$ . Prove that  $a \cdot (-1) = -a$ .

**Problem 3.** Let  $a, b \in \mathbb{F}$ . Prove that  $(-a) \cdot (-b) = a \cdot b$ .

**Problem 4.** Let  $a \in \mathbb{F}$ . Prove that  $-(-a) = a$ .

**Problem 5.** Let  $a, b \in \mathbb{F}$ . Prove that  $-(a + b) = (-a) + (-b)$ .

**Problem 6.** Let  $a, b, c \in \mathbb{F}$ . Prove that if  $a + b = a + c$ , then  $b = c$ .

## Part B Is It a Field?

**Problem 7.** For each of the following, determine whether it is a field under the given operations. If **yes**, give a brief justification. If **no**, state *exactly* which axiom fails and exhibit an explicit counterexample.

(a)  $(\mathbb{Z}, +, \cdot)$ : the integers with the usual operations.

(b)  $\mathbb{Z}/6\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}\}$  with addition and multiplication modulo 6.

(c) **[Challenge]**  $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\} \subset \mathbb{R}$ , with operations inherited from  $\mathbb{R}$ .

(d)  $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\} \subset \mathbb{C}$  (the *Gaussian integers*), with operations inherited from  $\mathbb{C}$ .

**Problem 8.** Consider the set  $\{0, 1, 2, 3\}$  of four elements, where 0 is the additive identity and 1 is the multiplicative identity. Define addition and multiplication operations on this set that make it a field.

- (a) Write down complete addition and multiplication tables for your field.
- (b) For each nonzero element, identify its multiplicative inverse from your table.
- (c) What is the characteristic of your field? Why does  $(\mathbb{Z}/4\mathbb{Z}, +, \cdot)$  *not* work, i.e., why is ordinary arithmetic modulo 4 not a field?

## Part C Challenge: Rational Functions

**Problem 9 (Challenge).** Let  $\mathbb{F}$  be any field. Define the set of *rational functions* over  $\mathbb{F}$ :

$$\mathbb{F}(X) = \left\{ \frac{p(X)}{q(X)} \mid p(X), q(X) \in \mathbb{F}[X], \quad q(X) \neq 0 \right\},$$

where we declare  $\frac{p}{q} = \frac{p'}{q'}$  whenever  $p \cdot q' = p' \cdot q$  in  $\mathbb{F}[X]$ , and define operations:

$$\frac{p}{q} + \frac{r}{s} = \frac{ps + rq}{qs}, \quad \frac{p}{q} \cdot \frac{r}{s} = \frac{pr}{qs}.$$

- (a) Identify the additive identity (“zero”) and the multiplicative identity (“one”) of  $\mathbb{F}(X)$ . Write down the additive inverse of  $p/q$  and the multiplicative inverse of a *nonzero* element  $p/q$ .
  
  
  
  
  
  
  
  
  
  
- (b) Verify axiom (F10): every nonzero element of  $\mathbb{F}(X)$  has a multiplicative inverse. What does it mean for  $p/q$  to be *nonzero*? Why does your inverse formula break down if  $p/q = 0$ ?
  
  
  
  
  
  
  
  
  
  
- (c) Axioms (F1)–(F9) are mostly inherited from  $\mathbb{F}[X]$ . Identify which of (F1)–(F9) requires the most care to verify directly (not just “inherited”), and explain in one or two sentences what you need to check. You do not need to write a full proof.
  
  
  
  
  
  
  
  
  
  
- (d) **(Bonus)** The operations above are *defined* on representatives  $p/q$ , so we must check they are *well-defined*. Show: if  $p/q = p'/q'$  and  $r/s = r'/s'$ , then  $p/q + r/s = p'/q' + r'/s'$ .

**Problem 10 (Harder Challenge: Characteristic).** For a field  $\mathbb{F}$ , define the *characteristic* as follows. For each integer  $n \geq 1$ , write

$$n \cdot 1_{\mathbb{F}} := \underbrace{1_{\mathbb{F}} + 1_{\mathbb{F}} + \cdots + 1_{\mathbb{F}}}_{n \text{ terms}}.$$

The characteristic  $\text{char}(\mathbb{F})$  is the smallest positive integer  $n$  such that  $n \cdot 1_{\mathbb{F}} = 0_{\mathbb{F}}$ , or 0 if no such  $n$  exists. (See Remark 1.1.18 in the notes.)

(a) Show that if  $\text{char}(\mathbb{F}) = n > 0$ , then  $n \cdot a = 0$  for *every*  $a \in \mathbb{F}$  (not just  $a = 1_{\mathbb{F}}$ ).

(b) Prove that  $\mathbb{F}$  has *no zero divisors*: if  $a \cdot b = 0$  then  $a = 0$  or  $b = 0$ .

(c) Prove that  $\text{char}(\mathbb{F})$  is either 0 or a prime number. That is, show there is no field whose characteristic is a composite number.

(d) Give an example of a field of characteristic 0 and a field of characteristic  $p$  for each prime  $p$ . For the latter, what is the smallest such field (by number of elements)?